



Homework 01
Data Structures and Algorithms, Fall 2026
Department of Software Engineering,
Faculty of Computer Science & Information Technology,
The Superior University Gold Campus, Lahore.
Submission Deadline: Sunday 11:59 PM, Sep 13, 2026

Question	CLO	Domian/BT Level	Total Marks
1	1,2	C2,C3	25

Advice and Submission Guidelines for Assignment

- Make sure that you read and understand each and every instruction. If you have any questions or comments you are encouraged to **discuss (only)** with your colleagues and instructor.
- Paste all codes and screenshots of output on word file, later you can convert this word file into PDF file as well. **Upload PDF file on the Google classroom** and keep code files with you (better in your own email) as it will be used for evaluation and viva.
- All the submitted evaluation instruments (quizzes, assignments, lab work, exams, and the project) will be checked for plagiarism.
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- Your word/pdf file should be **named as your complete roll number** e.g. bsse-f23-001.
- **Copied assignments will be graded as zero.**
- **Late Submission will not be accepted.**

While assignments are designed to assess your understanding and skills, they are also opportunities to learn and grow. Enjoy the process of solving problems and building solutions!

Question 1: Design the solution for following problems.

Marks: 25

a) Write a C++ program that:

- Declares an integer variable `num`.
- Stores a value entered by the user.
- Creates an integer pointer that stores the address of `num`.
- Displays:
 - The value of `num`
 - The address of `num`
 - The value stored in the pointer
 - The value accessed using the pointer

b) Write a C++ program that:

- Declares an integer variable `num`.

- Takes its value from the user.
- Creates a pointer pointing to `num`.
- Uses the pointer to change the value of `num`.
- Displays the value before and after modification.

c) Write a C++ program that:

- Declares two integer variables.
- Takes values from the user.
- Creates two pointers pointing to these variables.
- Uses the pointers to calculate and display:
 - Sum
 - Difference
 - Product

d) Write a C++ program that:

- Creates a 1D integer array of size 5.
- Takes 5 numbers from the user.
- Displays all the elements of the array.

e) Write a C++ program that:

- Creates a 1D integer array of size 5.
- Takes 5 numbers from the user.
- Finds and displays the **largest** and **smallest** element.

f) Write a C++ program that:

- Creates a 1D integer array.
- Takes 10 integers from the user.
- Counts how many numbers are **even** and how many are **odd**.
- Displays both counts.